



LABORATORY TEST REPORT

Client Name : Gerab System Technology LLC.

Address : Dubai - UAE.

Report No : WLR-ECI19-034

Sample No : WS-ECI19-034

Report Date : 20/03/2019

Introduction: Further to the request received from M/s. Gerab System Technology LLC dated 03rd March 2019, a sample of Fibre Optic Cable was tested using with IEC-60794-1-21 & 22.

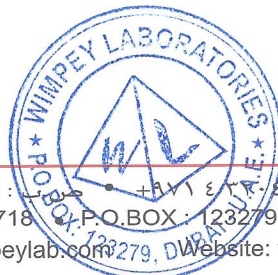
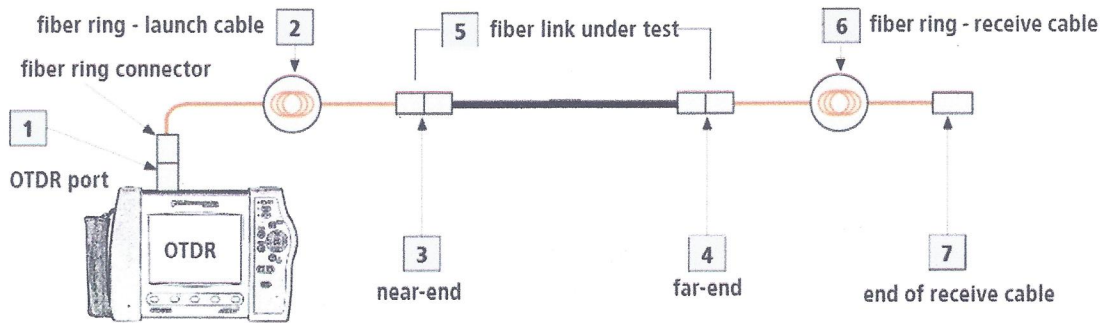
Sample Description : 4Fibre Single Mode LSZH for FTTH App G.6572
Manufacturer : Datacom
Model / Part No. : DC-FO-04-FTX097A-INZH
Test Method :

Testing Parameters	Test Method
Tensile Performance Test	IEC 60794-1-21 E1
Crush Test (Compression)	IEC 60794-1-21 E3A
Impact Resistance Test	IEC 60794-1-21 E4
Torsion Test	IEC 60794-1-21 E7
Bending Under Tension	IEC 60794-1-21 E18A
*Repeated Bending Test	IEC 60794-1-21 E6
Bend Test	IEC 60794-1-21 E11A
Temperature Cycling Test	IEC 60794-1-22 F1
Water Penetration Test	IEC 60794-1-22 F5
Outer Cable Dimension	ISO 13385-1

Sample received : 03/03/2019
Date of Test : 06/03/2019 – 19/03/2019
Tested by : SKC, SJV & VVS
Standard Test Condition : Temperature 23°C (±5°C)
(IEC 60794-1-20) Relative Humidity 20% To 70%

RESULTS

OTDR Connection diagram:





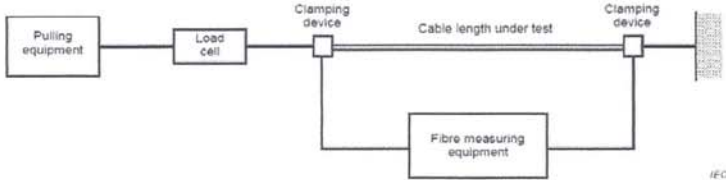
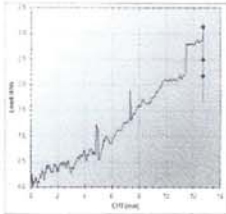
Description

Testing
Laboratory

Marking	DATACOM 4F 9/125 ITU-T 0657-A2 LSZH PN# DC-FO-04-FTX097A-INZH		
Sheath Color	Yellow		
Fiber Core color	Blue, Orange, Green, Brown		
No. of Loose Tube	N/A	No. of Ribbon per Tube	N/A
No. of Fiber per Tube	N/A	Total No. of Fiber	04

Attenuation Measurement		
Core Identification	Measured Attenuation(dB/Km)	
	1310nm	1550nm
Blue	0.316	0.182
Orange	0.327	0.194
Green	0.313	0.188
Brown	0.337	0.208

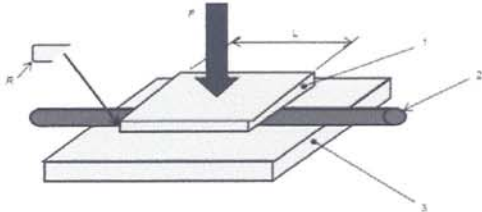
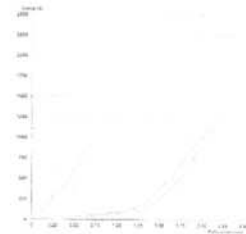
Tensile Performance Test: IEC 60794-1-21 E1

Tensile Performance Test				
		(Tensile Performance Graph) 		
Test Condition	Temperature (°C)		Relative Humidity (%)	
	22.8			
Sample Length (Meters)	Applied Force for 10 Min. Constant Long Term Load (N)			
	5080			
Result				
Identification (Loose Tube/Sheath Colour)	Fiber Color	Attenuation Change (dB) Max.		Result
		1310 nm	1550 nm	
Yellow	Blue	0.05	0.05	A.) Attenuation change and/or fiber strain not exceed the values given in the relevant specification. B.) No physical damage observed to the cable.

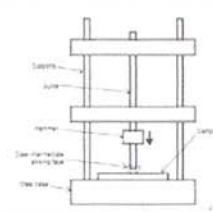


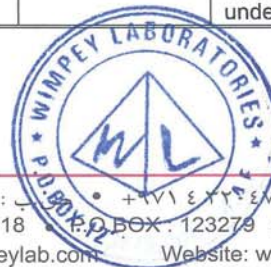


Crush Test: IEC 60794-1-21 E3A

Crush Test					Testing Laboratory
				(Crush Test Graph) 	
Test Condition	Temperature (°C)		Relative Humidity (%)		
	22.6		56		
Total Force for Applied (N/100mm)		Duration of Application of the Force (Short Term)	Number of Tests	Measured Spacing Between Test Places (under load) (mm)	
Short Term (N/100mm)	Long Term (N)				
1000	500	01 minute	03	1.90	
Result					
Identification (Loose Tube/Sheath Colour)	Fiber Color	Attenuation Change (dB) Max.		Result	
		1310 nm	1550 nm		
Yellow	Brown	0.05	0.03	A). No physical damage observed to the cable. B). No degradation of optical continuity. C). Measured loss of optical transmittance meets under the limit of standard specification	

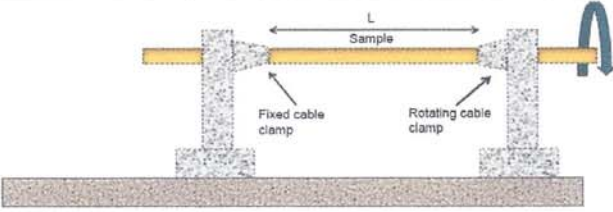
Impact Test: IEC 60794-1-21 E4

Impact Test				
				
Test Condition	Temperature (°C)		Relative Humidity (%)	
	22.8		55	
Number of Impacts	Impact Energy	Sample Length (Meters)	Frequency of Multiple Impacts	Location of Impacts on The Sample
05	1.0 J	25.0	05 Cycles	Top center of cable
Result				
Identification (Loose Tube/Sheath Colour)	Fiber Color	Attenuation Change (dB) Max.		Result
		1310 nm	1550 nm	
Yellow	Brown	0.03	0.02	A). No physical damage observed to the cable. B). No degradation of optical continuity. C). Measured loss of optical transmittance meets under the limit of standard specification.

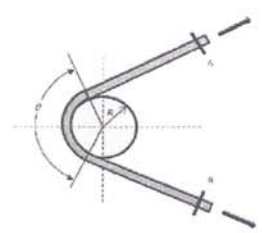


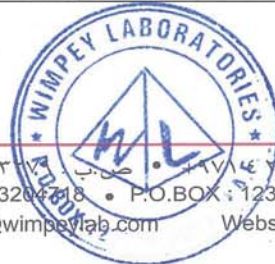


Torsion Test: IEC 60794-1-21 E7

Torsion Test							Testing Laboratory
							
Test Condition		Temperature (°C)		Relative Humidity (%)			
		22.9		56			
Test Length (Meters)	Bending Radius (mm)	Gauge Length (Meters)	No. of Cycles	Duration/ Cycles	Rotating Angle	Temp. of the specimen	
50.0	60	2.0	10 (4Part/Cycle)	1 Minute	180° Clock Wise & 180° Counter Clock Wise	Ambient Condition	
Result							
Identification (Loose Tube/Sheath Colour)	Fiber Color	Attenuation Change (dB) Max.		Result			
		1310 nm	1550 nm				
Yellow	Blue	0.07	0.06	A). No physical damage observed to the cable. B). No degradation of optical continuity. C). Measured loss of optical transmittance meets under the limit of standard specification.			

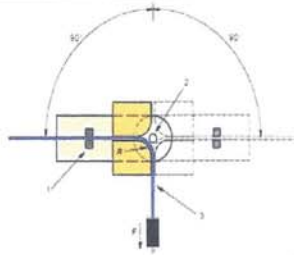
Bending Under Tension Test: IEC 60794-1-21 E18A

Bending Under Tension Test (Procedure – 01)				
				
Test Condition		Temperature (°C)		Relative Humidity (%)
		23.2		56
Max. Tension Applied(N)	Length of Sample (Meter)	Length Bend Under Tension (Meter)	Bend Angle Over Rollers	Tension Device
100	25.0	3.0 (Distance A-B)	180 °	Tensile Power Device
No. of Moving Cycles	Moving Speed	End Preparation	Radius (r) of rollers	Procedure
05	As per standard	As per standard	150 mm	01
Result				
Identification (Loose Tube/Sheath Colour)	Fiber Color	Attenuation Change (dB) Max.		Result
		1310 nm	1550 nm	
Yellow	Brown	0.06	0.06	A). No physical damage observed to the cable. B). No degradation of optical continuity. C). Measured loss of optical transmittance meets under the limit of standard specification.





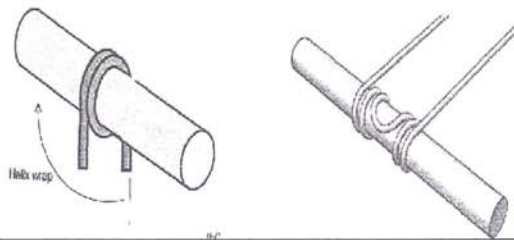
Repeated Bending Test



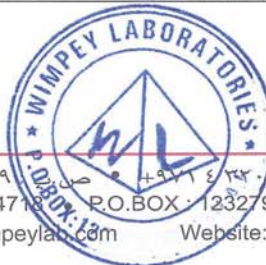
Test Condition	Temperature (°C)		Relative Humidity (%)	
	23.4		55	
Number of Cycles	Bending Radius (mm)	Sample Length (Meters)	Load (If applicable)	Precondition & Duration
20	30	5.0	Nil	Temp. : 23 °C RH. : 55% 24 Hours
Result				
Identification (Loose Tube/Sheath Colour)	Fiber Color	Attenuation Change (dB) Max.		Evaluation
		1310 nm	1550 nm	
Yellow	Brown	0.05	0.06	A). No physical damage observed to the cable. B). No degradation of optical continuity. C). Measured loss of optical transmittance meets under the limit of standard specification.

Bend Test: IEC 60794-1-21 E11A

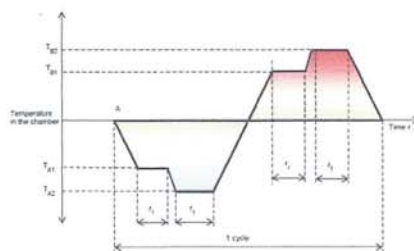
Bend Test (Procedure – 01)



Test Condition	Temperature (°C)		Relative Humidity (%)	
	23.2		56	
Test Mandrel Diameter (mm)		No. of Cycles	No. of Turns	Temp. of the specimen
30		3	4	Ambient Condition
Result				
Loose Tube Color	Fiber Color	Attenuation Change (dB) Max.		Evaluation
		1310 nm	1550 nm	
Yellow	Brown	0.05	0.06	A). No physical damage observed to the cable. B). No degradation of optical continuity. C). Measured loss of optical transmittance meets under the limit of standard specification.



Temperature Cycling



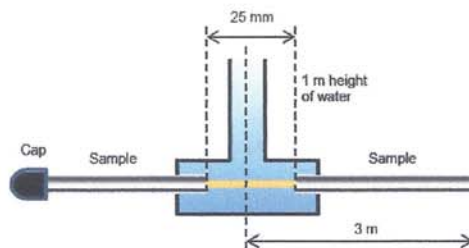
Test Condition	Temperature (°C)				Relative Humidity (%)		
	23.5				55		
Sample Length (Meters)	Temperature Limit	No. of Cycle	Type of Winding	Winding Diameter	Layer of Winding	Winding Tension (if any)	Min. Soak Duration per cycle
50.0	-10 to 70°C	02	Coil	300 mm	Multi-Layer	Nil	1.0 Hours

Result

Identification (Loose Tube/Sheath Colour)	Fiber Color	Attenuation Change (dB) Max.		Result
		1310 nm	1550 nm	
Yellow	Orange	0.05	0.06	A). No physical damage observed to the cable. B). No degradation of optical continuity. C). Measured loss of optical transmittance meets under the limit of standard specification.

Water Penetration Test : IEC 60794-1-22 F5A

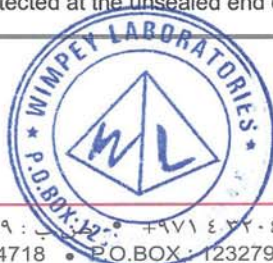
Water Penetration Test



Test Condition	Temperature (°C)			Relative Humidity (%)	
	23.1			57	
Sample Length (Meters)	Height of Water	Test Duration	Circumferential Portion of Sheath and Wrapping	Details of Dye (if used)	
3.0	1.0 Meter	24.0Hours	25 mm	Normal colors	

Result

Water not detected at the unsealed end of sample.





Dimension: ISO 13385-1

Outer Cable Dimension	
Description	Average Measured Value
Outer Diameter (Width x Height)	2.99 x 1.93 mm

Remarks:

The above tested parameters comply with the specification of IEC-60794-1-21 & 22.

* Non accredited scope by IAS.

Signed for and on behalf of Wimpey Laboratories

A. Umar Farook
Technical Manager

Test results relate only to the samples tested.

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